

DMS Database Design Document

Document Management System — Legal Subpoena & Medical Records Platform

Database Engine: MySQL 8.0+

Character Set: utf8mb4 / utf8mb4_unicode_ci

Storage Engine: InnoDB

Schema Name: dms_db

Version: 1.1

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1. Executive Summary

This document defines the relational database architecture for the **Document Management System (DMS)**, a legal/workers' compensation platform used to manage subpoenas, medical record requests, facility (law firm) accounts, provider serve targets, invoicing, reminders, and system audit trails.

The schema was reverse-engineered from the Next.js frontend application domain model. The application currently uses in-memory mock data; this design provides the production-ready MySQL foundation for backend API development.

Note: The companion SQL file `DMS_MySQL_Schema.sql` contains the executable CREATE TABLE statements. This document explains the design rationale, relationships, and field definitions.

Key Statistics

Metric	Value
Total Tables	22
Core Domains	Staff, Facilities, Providers, Orders, Invoicing, Audit/Auth
Foreign Key Relationships	32+
Primary Business Entity	orders (subpoena/case records)

2. System Overview & Domain Model

2.1 Business Context

DMS supports a records-retrieval workflow for workers' compensation (WCAB) legal cases:

1. **Facilities** (law firms) submit orders for medical/billing/employment/x-ray records.
2. **Orders** track applicant details, subpoena documents, serve targets, and workflow stages.
3. **Providers** (hospitals, clinics) are the custodians of records being subpoenaed.
4. **Matrix Employees** (internal DMS staff) process orders, create invoices, and manage facilities.
5. **Invoices** bill facilities for service fees, custodian fees, x-ray fees, and per-page charges.

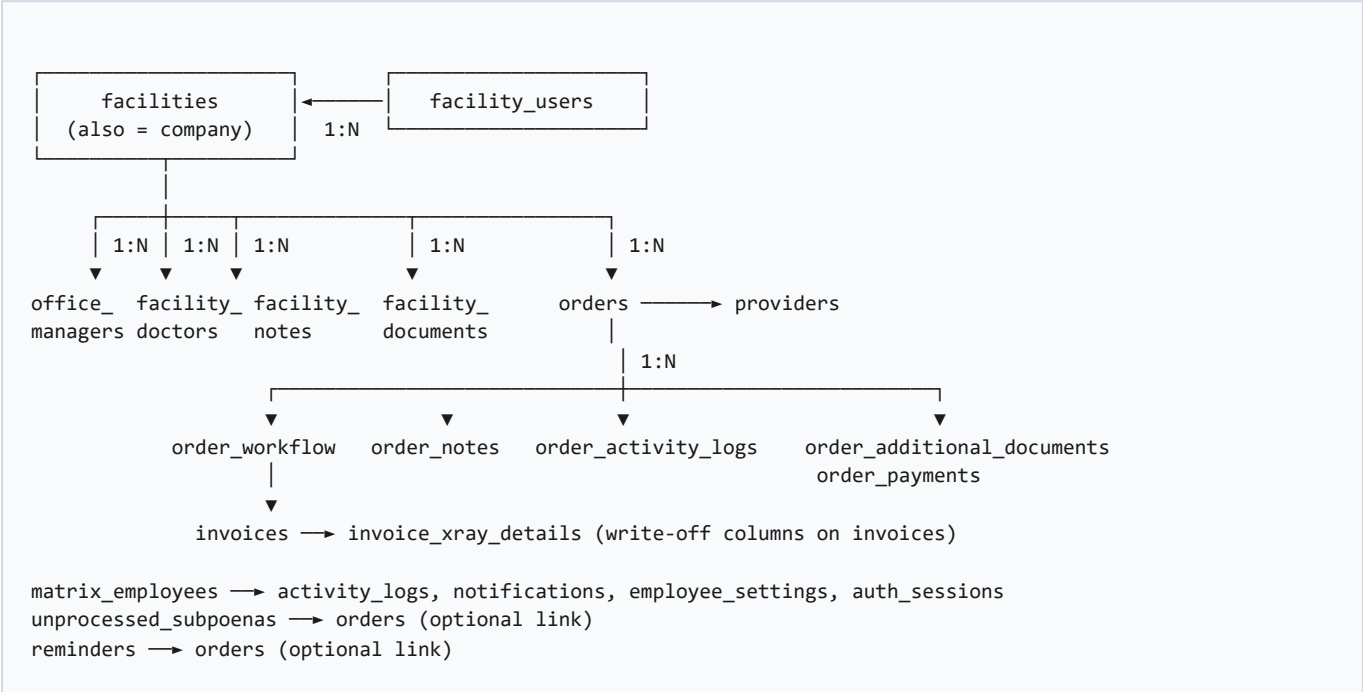
Facility = Company: The frontend treats each facility as its own billing company (company-wise invoices use the same names as the facilities list). There is no separate parent-company layer in the current UI.

2.2 Architectural Principles

- **Normalization:** Separate entities for facilities, providers, orders, payments, and invoices.
- **Referential Integrity:** All relationships enforced via InnoDB foreign keys.
- **Auditability:** Dedicated `activity_logs` and per-order activity tables.

- **Security:** Passwords stored as hashes; SSN stored encrypted with last-four display field.
- **Extensibility:** JSON column for facility user day-time schedules; workflow stages as rows.

3. Entity Relationship Overview



3.1 Relationship Summary

Parent Table	Child Table	Cardinality	On Delete
facilities	office_managers, facility_doctors, facility_notes, facility_documents, facility_users	1:N	CASCADE
facilities	orders	1:N	RESTRICT
providers	orders	1:N	SET NULL
orders	order_workflow_stages, order_payments, order_notes, order_activity_logs, order_additional_documents	1:N	CASCADE
orders	invoices	1:N	RESTRICT
invoices	invoice_xray_details	1:1	CASCADE
matrix_employees	notifications, employee_settings, auth_sessions	1:N	CASCADE
orders	unprocessed_subpoenas, reminders	1:0..1	SET NULL

4. Database Tables (Detailed)

4.1 Staff & Authentication

matrix_employees

Internal DMS operators who log into the admin dashboard.

Column	Type	Description
id	BIGINT UNSIGNED PK	Auto-increment primary key
name	VARCHAR(150)	Full display name
logon	VARCHAR(100) UNIQUE	Login username
email	VARCHAR(255) UNIQUE	Email address
password_hash	VARCHAR(255)	Bcrypt/Argon2 hash
role	ENUM	Manager Employee
last_login_at	DATETIME	Last successful login
terminated	TINYINT(1)	Soft-delete flag

auth_sessions

Active login sessions including 2FA verification state.

employee_settings

Per-employee notification preferences (new orders, invoice reminders, etc.).

4.2 Facilities (Customer Accounts)

facilities

Law firm / customer accounts that submit record requests. Also serves as the billing company in company-wise invoice reports.

Column	Type	Description
id	BIGINT UNSIGNED PK	Primary key
facility_name	VARCHAR(200)	Display name (maps to frontend facility and invoice company)
slug	VARCHAR(100) UNIQUE	URL/login identifier
user_name	VARCHAR(100) UNIQUE	Portal login
password_hash	VARCHAR(255)	Hashed password
contact_first/middle/last_name	VARCHAR	Primary contact person
address, zip_code, city, state	VARCHAR/CHAR	Mailing address
phone, fax, email	VARCHAR	Contact info
ip_addresses	TEXT	IP whitelist (newline-separated)

office_managers

Office manager contacts associated with a facility.

facility_doctors

Default/associated doctors per facility with active/default flags.

facility_notes

Internal notes on a facility (max 500 chars in UI).

facility_documents

Uploaded documents (Standard, Legal, Medical, Financial, Other).

facility_users

Portal users per facility with role, job title, and time-based access controls.

4.3 Providers

providers

Medical record custodians / serve targets (hospitals, clinics). Distinct from law-firm facilities.

4.4 Orders (Core Entity)

orders

Central subpoena/case record. Links facility (customer) to provider (serve target).

Column Group	Key Fields
Identity	order_number (unique), case_number, subpoena_ref, order_ref, court
Classification	order_type (medical billing employment xrays), status
Applicant	applicant names, aka, defendant, dob, ssn_encrypted, ssn_last_four
Injury	injury_type (specific cumulative)
Documents	subpoena_storage_path (on order); additional docs in order_additional_documents
Serve Target	serve_company_name, serve address/phone/fax/email
Contacts	contact1_*, contact2_* (name, title, phone, fax, email)
Dates	date_served, depo_due_date, delivery_date, subpoena_date, ready_date, invoice dates
Record Flags	flag_medical_records, flag_billing_records, flag_employment_records, flag_xrays, flag_other_record
CNR (Cert. No Records)	certificate_no_records, cnr_reason, cnr_delivery, cnr_date_sent, cnr_memo

order_workflow_stages

Tracks pipeline stages: Review Records → Serve → Custodian → SENT.

order_payments

Payment tracking for prepayment, custodian, and x-ray checks (one row per type per order).

order_notes / order_activity_logs

User notes and activity callbacks tied to orders.

order_additional_documents

Additional documents uploaded on the New Order form (Document Name + file). Supports multiple files per order.

Column	Type	Description
order_id	FK	Parent order
document_name	VARCHAR(255)	User-entered label
original_file_name	VARCHAR(255)	Uploaded filename
mime_type	VARCHAR(100)	application/pdf, image/jpeg, etc.
storage_path	VARCHAR(500)	File location after upload
file_size_bytes, page_count	BIGINT / INT	File metadata
uploaded_by, uploaded_at	FK / DATETIME	Upload audit

unprocessed_subpoenas

Uploaded subpoena PDFs in the unprocessed queue (/orders/unprocessed) awaiting linkage to an order.

Column	Type	Description
reference_code	VARCHAR(50) UNIQUE	Display ID e.g. SUB-001
file_name, storage_path	VARCHAR	File identity and location
mime_type	VARCHAR(100)	File MIME type
file_size_bytes, page_count	BIGINT / INT	File metadata
order_id	FK NULL	Linked order after processing
is_processed, processed_at	TINYINT / DATETIME	Processing state

reminders

Callback reminders with case number, applicant, assigned employee, and optional attachment.

4.5 Invoicing

invoices

Billing records with fee breakdowns, status tracking, and recipient emails.

Fee Fields	Type
served_amount, service_fee, custodian_fee, xray_fee	DECIMAL(12,2)
mileage, parking, other_fee	DECIMAL(12,2)
page_count, per_page_amount, total_amount	INT / DECIMAL
amount_paid, amount_due	DECIMAL(12,2)
writeoff_amount, writeoff_date, writeoff_by, writeoff_reason	DECIMAL / DATE / FK / TEXT
status	Pending Created Partial Needs Resend Unpaid Paid Written Off
invoice_type	Standard Insurance Bill Custodian Invoice X-Ray Edit

Write-off flow: When user clicks "Write Off" on invoice pages, set status = 'Written Off', record writeoff_amount (typically remaining amount_due), reduce amount_due, optionally set orders.status = 'Write Offs', and log in activity_logs.

invoice_xray_details

Extended x-ray invoice fields: exam date, view count, per-view amount, prepayment, check number.

4.6 Audit & Notifications

activity_logs

System-wide audit trail: action, module, company, performer, details.

notifications

In-app notifications for matrix employees (Order, Invoice, Reminder, Activity types).

5. Enumerations & Lookup Values

Enum Name	Values	Used In
Employee Role	Manager, Employee	matrix_employees.role
Facility Access Level	Admin, Manager, User	facility_users.access_level
Facility Job Title	Attorney, Partner, Office Manager, Legal Assistant, Paralegal, Records Clerk, ...	facility_users.job_title
Document Upload Type	Standard, Legal, Medical, Financial, Other	facility_documents.upload_type
Order Type	medical, billing, employment, xrays	orders.order_type
Order Status	Active, Ready, Ready to Pickup, Completed, Cancelled, No Subpoena, No Records, Write Offs	orders.status
Injury Type	specific, cumulative	orders.injury_type
CNR Delivery	email, fax, pickup	orders.cnr_delivery
Workflow Stage	Review Records, Serve, Custodian, SENT	order_workflow_stages.stage_name
Workflow Status	pending, complete, failed, sent	order_workflow_stages.stage_status
Payment Type	prepayment, custodian, xray	order_payments.payment_type
Invoice Status	Pending, Created, Partial, Needs Resend, Unpaid, Paid, Written Off	invoices.status
Invoice Type	Standard, Insurance Bill, Custodian Invoice, X-Ray, Edit	invoices.invoice_type
Activity Module	Orders, Billing, Employees, Processing, Reports, Security	activity_logs.module
Notification Type	Order, Invoice, Reminder, Activity	notifications.notification_type

Derived Values (Not Stored)

- **Rush Level:** Rush 1 / Rush 2 / Rush 3 — computed from days since subpoena_date.
- **Invoice Age (days):** Computed from invoice_date vs current date.

6. Indexing & Performance Strategy

Table	Index	Purpose
orders	order_number (UNIQUE), status, case_number, subpoena_date, applicant names	Order search, filtering, rush reports
facilities	slug (UNIQUE), city+state	Lookup by slug, geographic reports
invoices	status, invoice_date, facility_id, writeoff_by	Billing reports, company-wise invoices, write-offs
order_additional_documents	order_id	Load additional docs per order
unprocessed_subpoenas	is_processed, uploaded_at, order_id	Unprocessed subpoena queue
activity_logs	log_date+log_time, module	Activity report queries
reminders	callback_date, case_number	Due reminder dashboards
notifications	employee_id+is_read	Unread notification badge

7. Security & Compliance Considerations

- **PII Protection:** SSN stored in `ssn_encrypted` (VARBINARY); only `ssn_last_four` displayed in UI.
- **Password Storage:** All passwords use one-way hashing (bcrypt recommended, min cost 12).
- **File Storage:** Document paths stored in DB; actual files in secure object storage (S3/Azure Blob).
- **IP Whitelisting:** Facility `ip_addresses` field supports access restriction.
- **Time-Based Access:** Facility users have authorized hours and per-day schedules.
- **Audit Trail:** All critical actions logged in `activity_logs` with performer identity.
- **HIPAA Awareness:** Medical record metadata requires encryption at rest and in transit.

8. MySQL Table Creation Scripts

Execute the following script to create the complete database schema:

```

-- Create database
CREATE DATABASE IF NOT EXISTS dms_db
  CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;
USE dms_db;

-- 1. STAFF
CREATE TABLE matrix_employees (
  id BIGINT UNSIGNED NOT NULL AUTO_INCREMENT,
  name VARCHAR(150) NOT NULL,
  logon VARCHAR(100) NOT NULL,
  email VARCHAR(255) NOT NULL,
  password_hash VARCHAR(255) NOT NULL,
  role ENUM('Manager','Employee') NOT NULL DEFAULT 'Employee',
  last_login_at DATETIME NULL,
  terminated TINYINT(1) NOT NULL DEFAULT 0,
  created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
  updated_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
  PRIMARY KEY (id),
  UNIQUE KEY uq_matrix_employees_logon (logon),
  UNIQUE KEY uq_matrix_employees_email (email)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

-- 2. FACILITIES (also used as billing company)
CREATE TABLE facilities (
  id BIGINT UNSIGNED NOT NULL AUTO_INCREMENT,
  facility_name VARCHAR(200) NOT NULL,
  slug VARCHAR(100) NULL,
  user_name VARCHAR(100) NOT NULL,
  password_hash VARCHAR(255) NOT NULL,
  contact_first_name VARCHAR(100) NULL,
  contact_middle_name VARCHAR(100) NULL,
  contact_last_name VARCHAR(100) NULL,
  address VARCHAR(255) NULL,
  zip_code CHAR(5) NULL,
  city VARCHAR(100) NULL,
  state CHAR(2) NULL,
  phone VARCHAR(20) NULL,
  fax VARCHAR(20) NULL,
  email VARCHAR(255) NOT NULL,
  ip_addresses TEXT NULL,
  is_active TINYINT(1) NOT NULL DEFAULT 1,
  created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
  updated_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
  PRIMARY KEY (id),
  UNIQUE KEY uq_facilities_user_name (user_name),
  UNIQUE KEY uq_facilities_slug (slug)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

CREATE TABLE office_managers (
  id BIGINT UNSIGNED NOT NULL AUTO_INCREMENT,
  facility_id BIGINT UNSIGNED NOT NULL,
  first_name VARCHAR(100) NOT NULL,
  middle_name VARCHAR(100) NULL,
  last_name VARCHAR(100) NOT NULL,
  phone VARCHAR(20) NULL,
  email VARCHAR(255) NULL,
  created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
  updated_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
  PRIMARY KEY (id),
  CONSTRAINT fk_office_managers_facility
    FOREIGN KEY (facility_id) REFERENCES facilities(id)
    ON DELETE CASCADE ON UPDATE CASCADE

```

```
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;
```

```
CREATE TABLE facility_doctors (  
  id BIGINT UNSIGNED NOT NULL AUTO_INCREMENT,  
  facility_id BIGINT UNSIGNED NOT NULL,  
  office_name VARCHAR(200) NULL,  
  first_name VARCHAR(100) NULL,  
  middle_name VARCHAR(100) NULL,  
  last_name VARCHAR(100) NULL,  
  phone VARCHAR(20) NULL,  
  fax VARCHAR(20) NULL,  
  email VARCHAR(255) NULL,  
  is_default TINYINT(1) NOT NULL DEFAULT 0,  
  is_active TINYINT(1) NOT NULL DEFAULT 1,  
  created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,  
  updated_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,  
  PRIMARY KEY (id),  
  CONSTRAINT fk_facility_doctors_facility  
    FOREIGN KEY (facility_id) REFERENCES facilities(id)  
    ON DELETE CASCADE ON UPDATE CASCADE  
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;
```

```
CREATE TABLE facility_notes (  
  id BIGINT UNSIGNED NOT NULL AUTO_INCREMENT,  
  facility_id BIGINT UNSIGNED NOT NULL,  
  note_date DATE NOT NULL,  
  created_by BIGINT UNSIGNED NULL,  
  author_name VARCHAR(150) NOT NULL,  
  note VARCHAR(500) NOT NULL,  
  created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,  
  updated_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,  
  PRIMARY KEY (id),  
  CONSTRAINT fk_facility_notes_facility  
    FOREIGN KEY (facility_id) REFERENCES facilities(id) ON DELETE CASCADE,  
  CONSTRAINT fk_facility_notes_created_by  
    FOREIGN KEY (created_by) REFERENCES matrix_employees(id) ON DELETE SET NULL  
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;
```

```
CREATE TABLE facility_documents (  
  id BIGINT UNSIGNED NOT NULL AUTO_INCREMENT,  
  facility_id BIGINT UNSIGNED NOT NULL,  
  document_name VARCHAR(255) NOT NULL,  
  upload_type ENUM('Standard','Legal','Medical','Financial','Other') NOT NULL DEFAULT 'Standard',  
  file_type VARCHAR(50) NULL,  
  storage_path VARCHAR(500) NOT NULL,  
  file_size_bytes BIGINT UNSIGNED NULL,  
  uploaded_by BIGINT UNSIGNED NULL,  
  uploaded_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,  
  created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,  
  updated_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,  
  PRIMARY KEY (id),  
  CONSTRAINT fk_facility_documents_facility  
    FOREIGN KEY (facility_id) REFERENCES facilities(id) ON DELETE CASCADE,  
  CONSTRAINT fk_facility_documents_uploaded_by  
    FOREIGN KEY (uploaded_by) REFERENCES matrix_employees(id) ON DELETE SET NULL  
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;
```

```
CREATE TABLE facility_users (  
  id BIGINT UNSIGNED NOT NULL AUTO_INCREMENT,  
  facility_id BIGINT UNSIGNED NOT NULL,  
  first_name VARCHAR(100) NOT NULL,  
  last_name VARCHAR(100) NOT NULL,
```

```

email VARCHAR(255) NOT NULL,
user_logon VARCHAR(100) NOT NULL,
password_hash VARCHAR(255) NOT NULL,
access_level ENUM('Admin','Manager','User') NOT NULL DEFAULT 'User',
job_title ENUM('Attorney','Partner','Office Manager','Legal Assistant',
'Paralegal','Records Clerk','Case Coordinator','Trust Manager',
'CPA','Medical Director','Other') NOT NULL DEFAULT 'Other',
authorized_from TIME NULL,
authorized_through TIME NULL,
access_mon TINYINT(1) NOT NULL DEFAULT 1,
access_tue TINYINT(1) NOT NULL DEFAULT 1,
access_wed TINYINT(1) NOT NULL DEFAULT 1,
access_thu TINYINT(1) NOT NULL DEFAULT 1,
access_fri TINYINT(1) NOT NULL DEFAULT 1,
access_sat TINYINT(1) NOT NULL DEFAULT 0,
access_sun TINYINT(1) NOT NULL DEFAULT 0,
day_times_json JSON NULL,
is_active TINYINT(1) NOT NULL DEFAULT 1,
created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
updated_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
PRIMARY KEY (id),
UNIQUE KEY uq_facility_users_logon (facility_id, user_logon),
CONSTRAINT fk_facility_users_facility
FOREIGN KEY (facility_id) REFERENCES facilities(id) ON DELETE CASCADE
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

```

Full Script: Tables 3–6 (providers, orders, invoicing, audit/auth) are included in the complete file database/DMS_MySQL_Schema.sql (22 tables total). Run: `mysql -u root -p < DMS_MySQL_Schema.sql`

-- 3. PROVIDERS

```
CREATE TABLE providers (  
  id BIGINT UNSIGNED NOT NULL AUTO_INCREMENT,  
  company_name VARCHAR(255) NOT NULL,  
  address VARCHAR(255) NULL, zip_code CHAR(5) NULL,  
  city VARCHAR(100) NULL, state CHAR(2) NULL,  
  phone VARCHAR(20) NULL, fax VARCHAR(20) NULL,  
  email VARCHAR(255) NULL,  
  is_active TINYINT(1) NOT NULL DEFAULT 1,  
  created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,  
  updated_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,  
  PRIMARY KEY (id)  
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;
```

-- 4. ORDERS (abbreviated – see full SQL file for all columns)

```
CREATE TABLE orders (  
  id BIGINT UNSIGNED NOT NULL AUTO_INCREMENT,  
  order_number VARCHAR(50) NOT NULL,  
  facility_id BIGINT UNSIGNED NOT NULL,  
  provider_id BIGINT UNSIGNED NULL,  
  order_type ENUM('medical','billing','employment','xrays') NOT NULL,  
  status ENUM('Active','Ready','Ready to Pickup','Completed','Cancelled',  
    'No Subpoena','No Records','Write Offs') NOT NULL DEFAULT 'Active',  
  case_number VARCHAR(50) NULL,  
  applicant_first_name VARCHAR(100) NULL,  
  applicant_last_name VARCHAR(100) NULL,  
  subpoena_date DATE NULL,  
  certificate_no_records TINYINT(1) NOT NULL DEFAULT 0,  
  created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,  
  updated_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,  
  PRIMARY KEY (id),  
  UNIQUE KEY uq_orders_order_number (order_number),  
  CONSTRAINT fk_orders_facility FOREIGN KEY (facility_id)  
    REFERENCES facilities(id) ON DELETE RESTRICT,  
  CONSTRAINT fk_orders_provider FOREIGN KEY (provider_id)  
    REFERENCES providers(id) ON DELETE SET NULL  
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;
```

-- 4b. ORDER ADDITIONAL DOCUMENTS

```
CREATE TABLE order_additional_documents (  
  id BIGINT UNSIGNED NOT NULL AUTO_INCREMENT,  
  order_id BIGINT UNSIGNED NOT NULL,  
  document_name VARCHAR(255) NOT NULL,  
  original_file_name VARCHAR(255) NOT NULL,  
  mime_type VARCHAR(100) NULL,  
  storage_path VARCHAR(500) NOT NULL,  
  file_size_bytes BIGINT UNSIGNED NULL,  
  page_count INT UNSIGNED NULL,  
  uploaded_by BIGINT UNSIGNED NULL,  
  uploaded_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,  
  created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,  
  updated_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,  
  PRIMARY KEY (id),  
  CONSTRAINT fk_order_additional_documents_order  
    FOREIGN KEY (order_id) REFERENCES orders(id) ON DELETE CASCADE  
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;
```

-- 4c. UNPROCESSED SUBPOENAS

```
CREATE TABLE unprocessed_subpoenas (  
  id BIGINT UNSIGNED NOT NULL AUTO_INCREMENT,  
  reference_code VARCHAR(50) NOT NULL,  
  file_name VARCHAR(255) NOT NULL,
```



```

storage_path VARCHAR(500) NOT NULL,
mime_type VARCHAR(100) NULL,
file_size_bytes BIGINT UNSIGNED NULL,
page_count INT UNSIGNED NULL,
order_id BIGINT UNSIGNED NULL,
is_processed TINYINT(1) NOT NULL DEFAULT 0,
processed_at DATETIME NULL,
uploaded_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
PRIMARY KEY (id),
UNIQUE KEY uq_unprocessed_subpoenas_ref (reference_code)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

```

-- 5. INVOICES (with write-off support)

```

CREATE TABLE invoices (
  id BIGINT UNSIGNED NOT NULL AUTO_INCREMENT,
  order_id BIGINT UNSIGNED NOT NULL,
  facility_id BIGINT UNSIGNED NOT NULL,
  invoice_type ENUM('Standard','Insurance Bill','Custodian Invoice','X-Ray','Edit') NOT NULL,
  status ENUM('Pending','Created','Partial','Needs Resend','Unpaid','Paid','Written Off') NOT NULL,
  total_amount DECIMAL(12,2) NOT NULL DEFAULT 0.00,
  amount_paid DECIMAL(12,2) NOT NULL DEFAULT 0.00,
  amount_due DECIMAL(12,2) NOT NULL DEFAULT 0.00,
  writeoff_amount DECIMAL(12,2) NOT NULL DEFAULT 0.00,
  writeoff_date DATE NULL,
  writeoff_by BIGINT UNSIGNED NULL,
  writeoff_reason TEXT NULL,
  created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
  updated_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
  PRIMARY KEY (id),
  CONSTRAINT fk_invoices_order FOREIGN KEY (order_id)
    REFERENCES orders(id) ON DELETE RESTRICT,
  CONSTRAINT fk_invoices_facility FOREIGN KEY (facility_id)
    REFERENCES facilities(id) ON DELETE RESTRICT,
  CONSTRAINT fk_invoices_writeoff_by FOREIGN KEY (writeoff_by)
    REFERENCES matrix_employees(id) ON DELETE SET NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

```

-- 6. ACTIVITY LOGS

```

CREATE TABLE activity_logs (
  id BIGINT UNSIGNED NOT NULL AUTO_INCREMENT,
  log_date DATE NOT NULL,
  log_time TIME NOT NULL,
  action VARCHAR(100) NOT NULL,
  module ENUM('Orders','Billing','Employees','Processing','Reports','Security') NOT NULL,
  performer_name VARCHAR(150) NOT NULL,
  details TEXT NULL,
  created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
  PRIMARY KEY (id)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

```

Complete Table List

#	Table Name	Domain
1	matrix_employees	Staff
2	facilities	Facilities / Billing company
3	office_managers	Facilities
4	facility_doctors	Facilities
5	facility_notes	Facilities
6	facility_documents	Facilities
7	facility_users	Facilities
8	providers	Providers
9	orders	Orders
10	order_workflow_stages	Orders
11	order_payments	Orders
12	order_notes	Orders
13	order_activity_logs	Orders
14	order_additional_documents	Orders
15	unprocessed_subpoenas	Orders
16	reminders	Orders
17	invoices	Invoicing / Write-offs
18	invoice_xray_details	Invoicing
19	activity_logs	Audit
20	notifications	Audit
21	employee_settings	Auth
22	auth_sessions	Auth